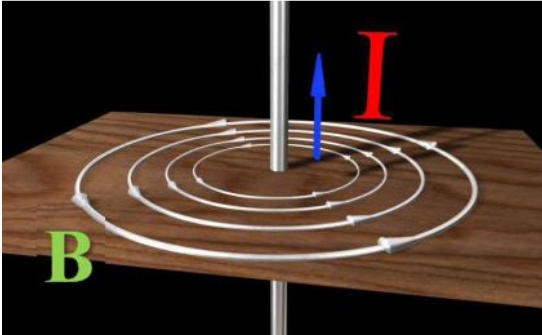
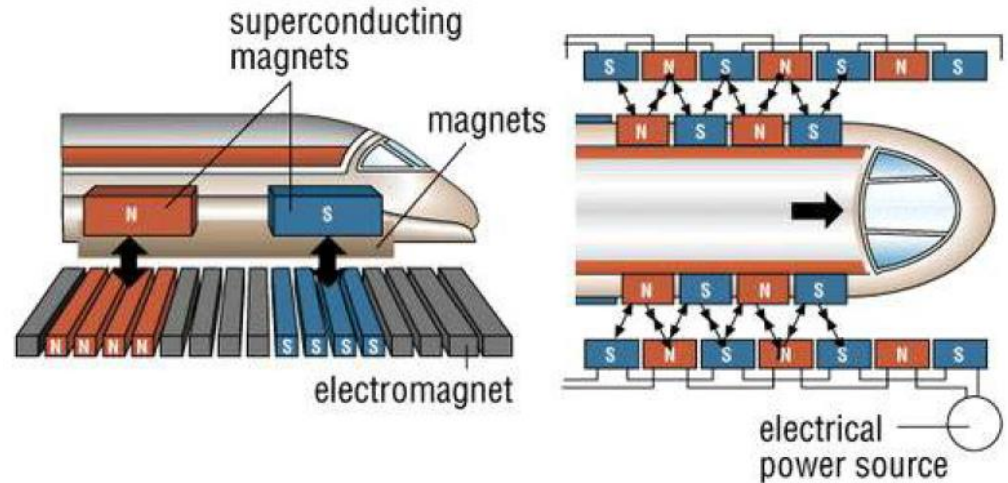


Ch29 Magnetic field due to currents

B-field around a wire with current



Maglev Trains

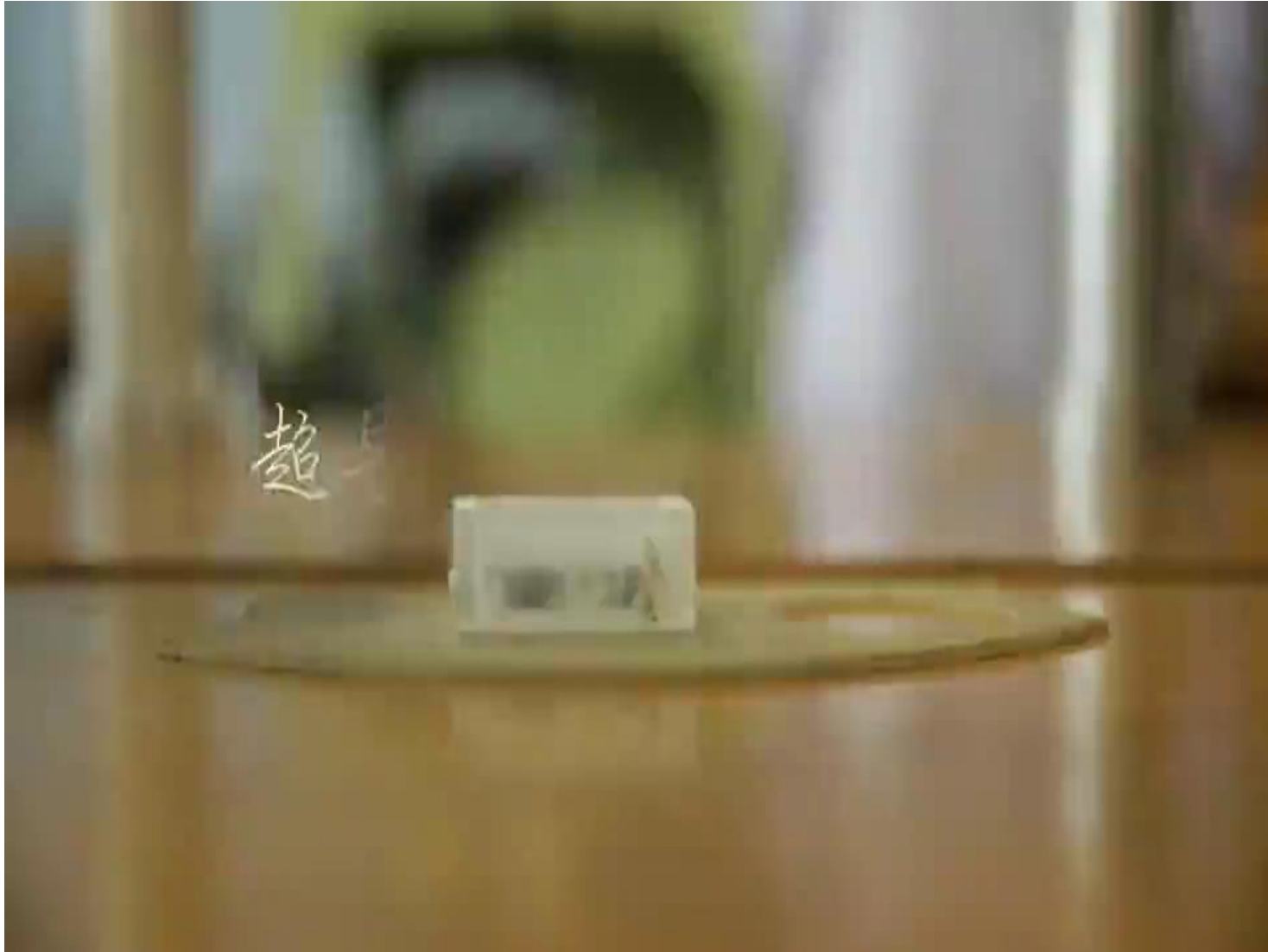


Electromagnet



<https://haokan.baidu.com/v?vid=10856230734533612165&pd=bjh&fr=bjhaauthor&type=video>
<https://haokan.baidu.com/v?vid=13960662013054252333>

Superconducting Maglev Trains



Ch29 Magnetic field due to currents

- Magnetic Fields Due to A Current

Biot–Savart (“Leo and bazaar”) Law

- Force Between Two Parallel Currents
- Ampere’s Law
- Solenoids and Toroids
- A Current-Carrying Coil As A Magnetic Dipole

<https://haokan.baidu.com/v?vid=11522194411454758837>

Biot-Savart Law (毕奥—萨伐尔定律)



Jean-Baptiste Biot, (1774-1862), French physicist, astronomer, and mathematician who helped formulate the Biot-Savart law

Félix Savart (1791-1841) was a French physicist, best known for the Biot-Savart law. He gave his name to the savart, a unit of measurement for musical intervals



Biot-Savart Law (毕奥—萨伐尔定律)

The electric current generates a magnetic field according to the Biot-Savart law.

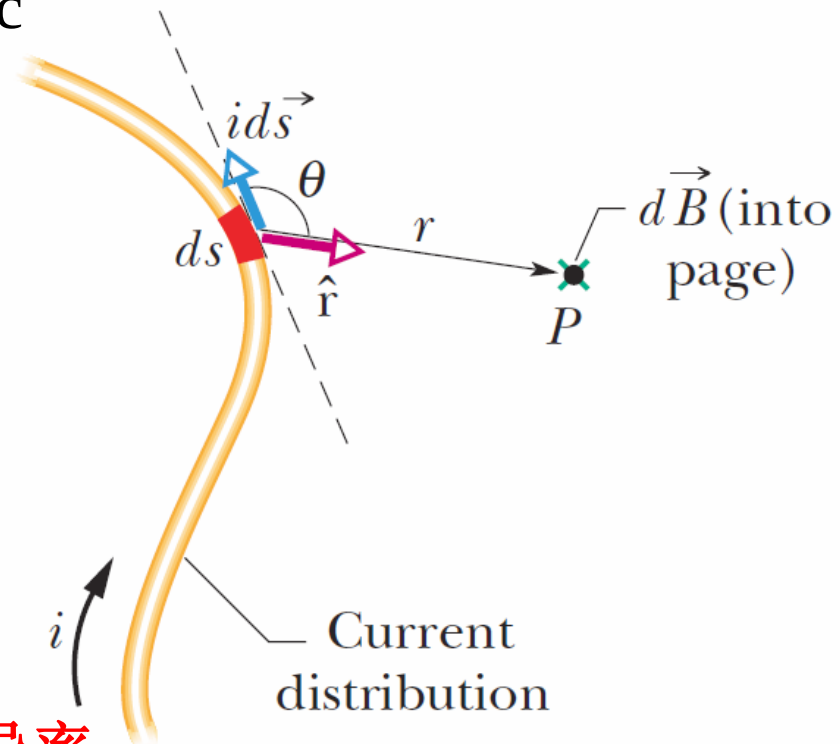
The *magnitude* of the field dB produced at a point P by a current-length element ids is

$$dB = \frac{\mu_0}{4\pi} \frac{i ds \sin \theta}{r^2}$$

or in vector form,

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2} \quad (\text{Biot-Savart law})$$

μ_0 – 真空磁导率;
 i – 导线中流过电流



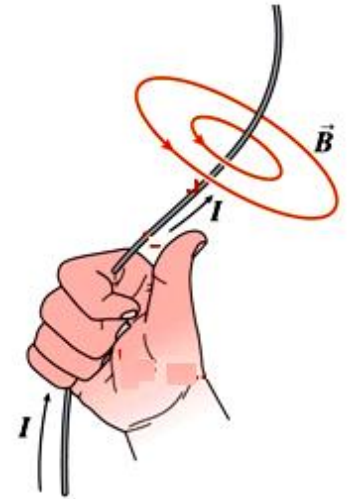
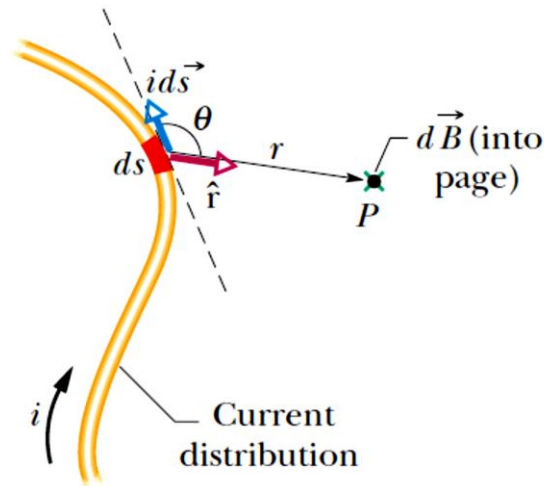
$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$$

Biot-Savart law and Coulomb's law

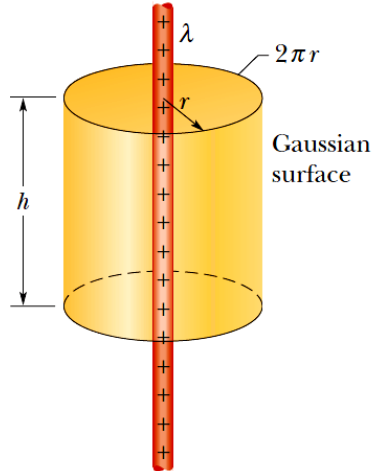
	Biot-Savart Law	Coulomb's Law
Expression	$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{id\mathbf{l} \times \mathbf{e}_r}{r^2}$	$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dq\mathbf{e}_r}{r^2}$
Inversed square law?	Yes	Yes
Infinitesimal element	$id\mathbf{l}$ (vector)	dq (scalar)
Coefficient	$\frac{\mu_0}{4\pi}$	$\frac{1}{4\pi\epsilon_0}$
Operation	Cross product	Scalar product

B-field of a current carrying wire

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s}}{r^2} \times \hat{r}$$



For E-field of a straight wire with linear charge distribution



$$E(2\pi rh) = \frac{Q_{enc}}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

For B-field of a wire with steady current:

$$B = \frac{\mu_0 i}{2\pi R}$$

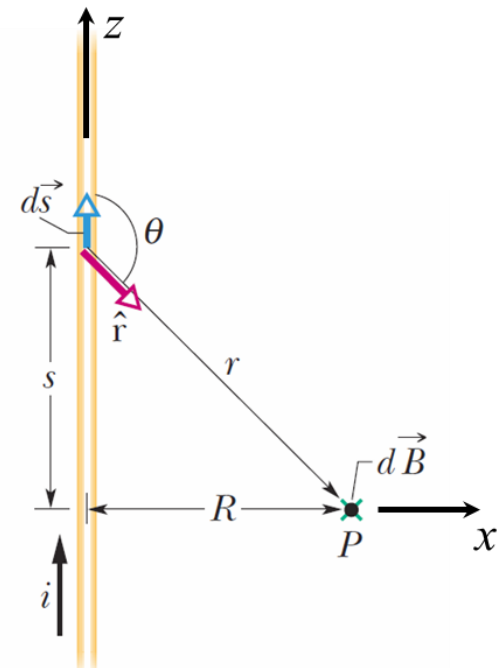
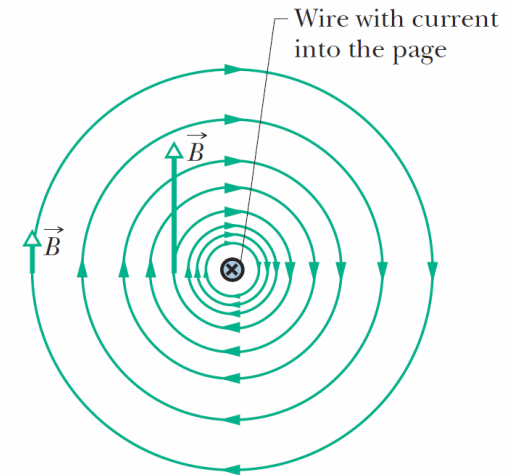
1) By integration

2) By Gauss' Law

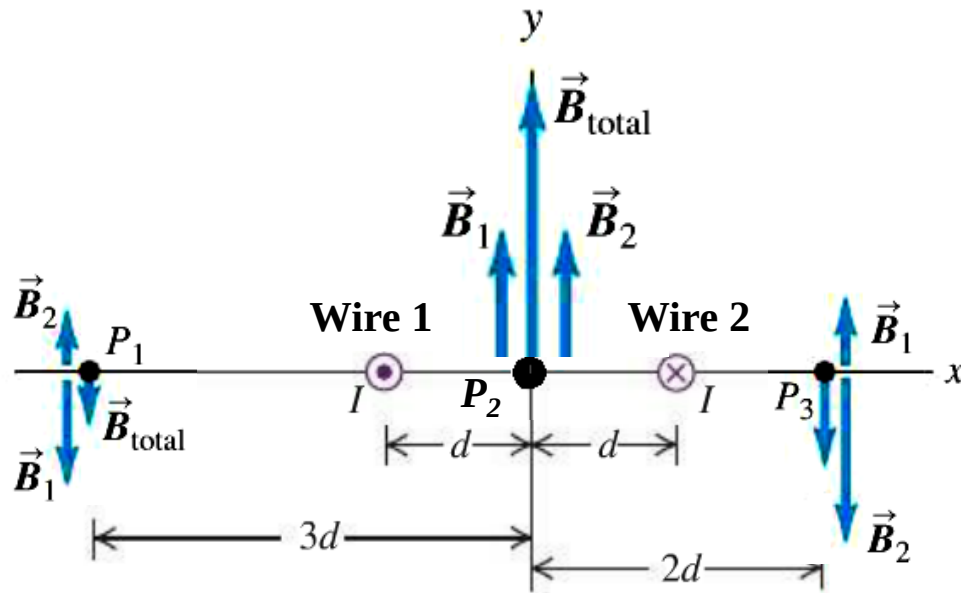
Magnetic field due to a straight wire of current

To calculate the B -field generated by a long, straight wire of current, we need to integrate $d\mathbf{B}$ from $-\infty$ to $+\infty$, in the mean time respecting the cross product:

$$\begin{aligned}
 \mathbf{B} &= \int d\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{id\mathbf{l} \times \mathbf{e}_r}{r^2} \\
 &= \frac{\mu_0 i}{4\pi} 2 \int_0^\infty \frac{dl \mathbf{e}_z \times \mathbf{e}_r}{R^2 + l^2} = \frac{\mu_0 i}{2\pi} \int_0^\infty \frac{dl \sin \theta \mathbf{e}_y}{R^2 + l^2} \\
 &= \frac{\mu_0 i}{2\pi} \int_0^\infty \frac{dl R \mathbf{e}_y}{(R^2 + l^2)^{3/2}} \quad \left[\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2(x^2 + a^2)^{1/2}} \right] \\
 &= \frac{\mu_0 i}{2\pi R} \mathbf{e}_y \left[\frac{l}{\sqrt{l^2 + R^2}} \right]_0^\infty = \frac{\mu_0 i}{2\pi R} \mathbf{e}_y.
 \end{aligned}$$



Example – Magnetic field of two wires



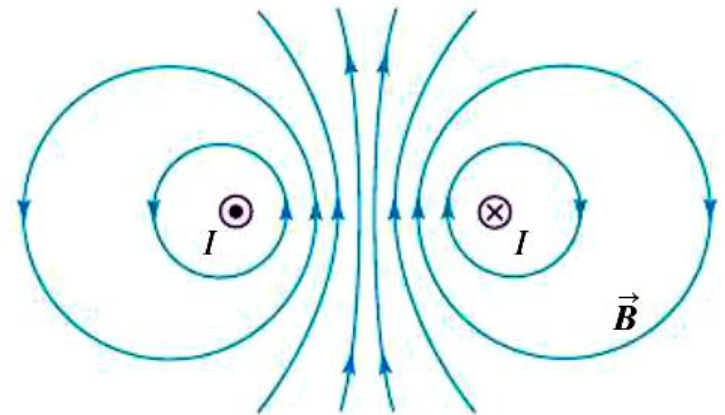
B-field of a wire with steady current:

$$B = \frac{\mu_0 i}{2\pi R}$$

$$\text{At } P_2, B_{\text{total}} = B_1 + B_2 = \frac{\mu_0 I}{2\pi d} + \frac{\mu_0 I}{2\pi d} = \frac{\mu_0 I}{\pi d}$$

$$\text{At } P_1, B_{\text{total}} = -\frac{\mu_0 I}{4\pi d} + \frac{\mu_0 I}{8\pi d} = -\frac{\mu_0 I}{8\pi d}$$

$$\text{At } P_3, B_{\text{total}} = \frac{\mu_0 I}{6\pi d} - \frac{\mu_0 I}{2\pi d} = -\frac{\mu_0 I}{3\pi d}$$

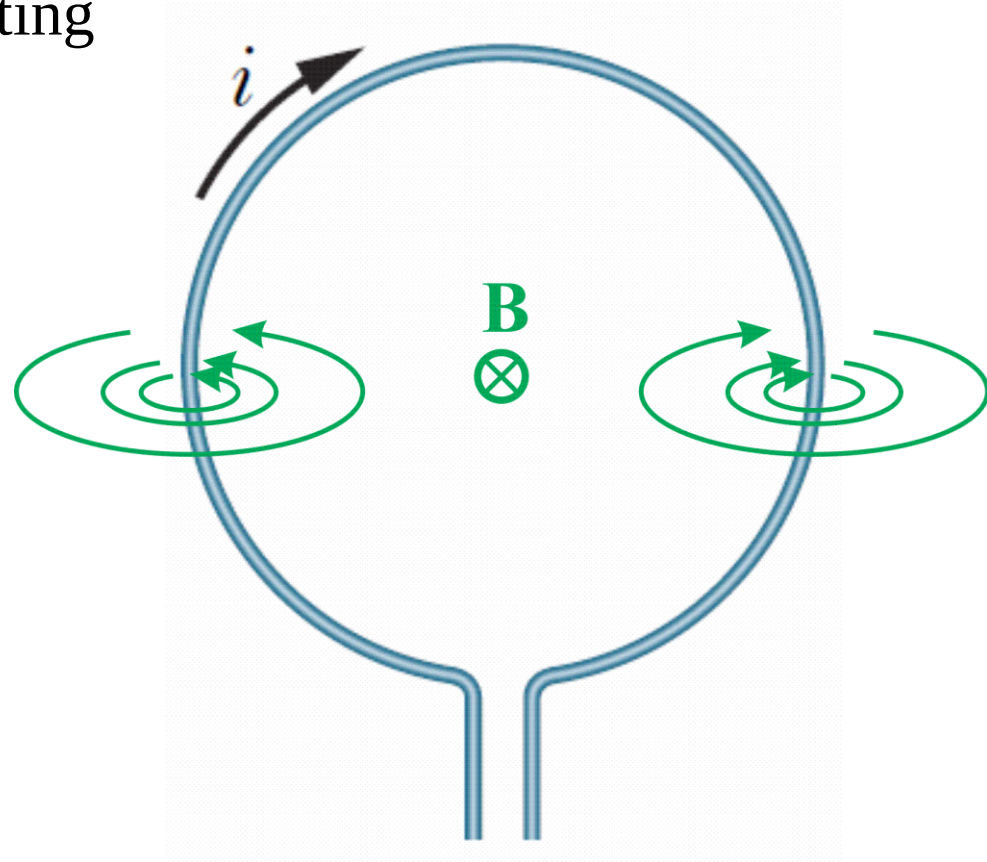


Magnetic field due to a ring of current

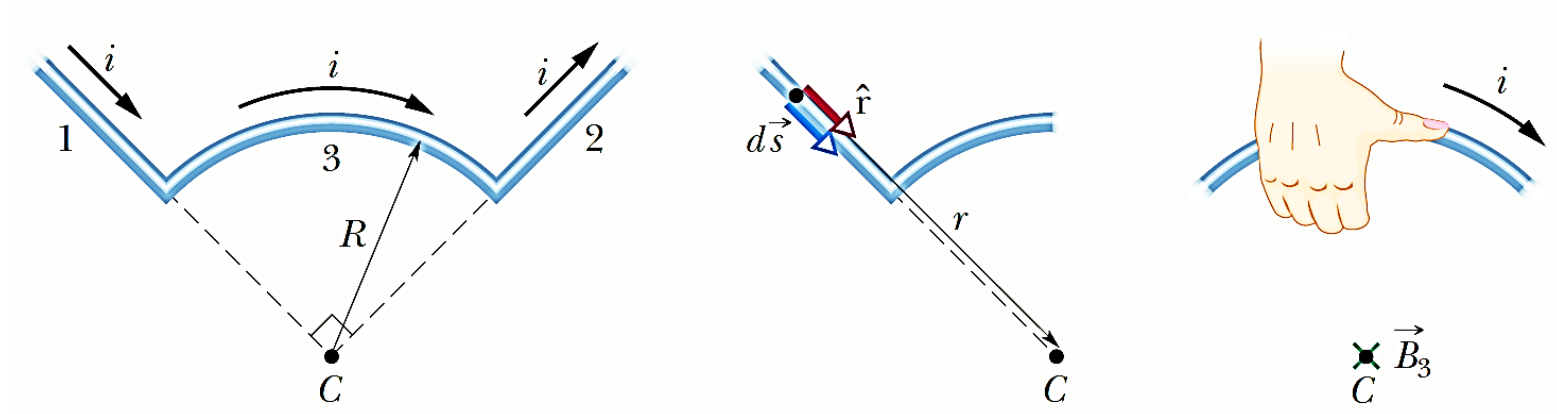
A ring of clockwise flowing current i , radius R produces an inward pointing B -field at the center of the ring.

According to Biot-Savart law, the magnitude of the B -field at the center is

$$B = \int dB = \frac{\mu_0}{4\pi} \int \frac{idl}{r^2}$$
$$= \frac{\mu_0 i}{4\pi R^2} \int_0^{2\pi R} dl = \frac{\mu_0 i}{2R}.$$



An example



Calculate the B -field at point C due to the current as shown. For Sections 1 & 2, $d\mathbf{l}$ is parallel (antiparallel) to $\mathbf{e}_r$. So

$$\mathbf{B}_1 = \mathbf{B}_2 = \frac{\mu_0}{4\pi} \int \frac{id\mathbf{l} \mathbf{e}_r \times \mathbf{e}_r}{r^2} = 0.$$

For Section 3, according to the right figure, the field is pointing inside the page. Its magnitude is

$$B_3 = \frac{\mu_0 i}{4\pi R^2} \int_0^{\pi R/2} dl = \frac{\mu_0 i}{8R}.$$

Force between two parallel currents

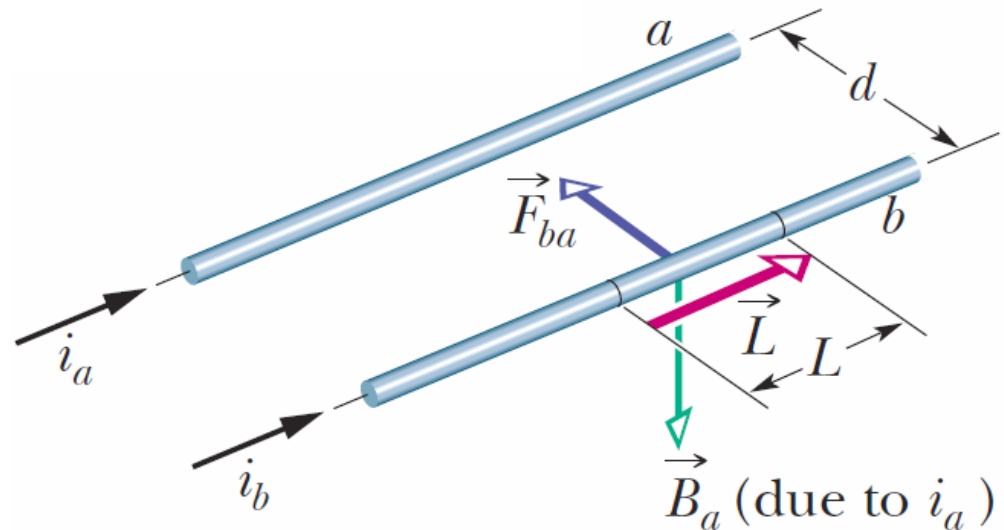
Due to the B -field created by the first wire (long, straight), a force is exerted on a second wire of current parallel to the first wire.

From previous slices, the B -field created by current a at current b is pointing downward, with magnitude,

For a section of length L at b , this gives rise to a force

$$F = i_b L B = \frac{\mu_0 L i_a i_b}{2\pi d},$$

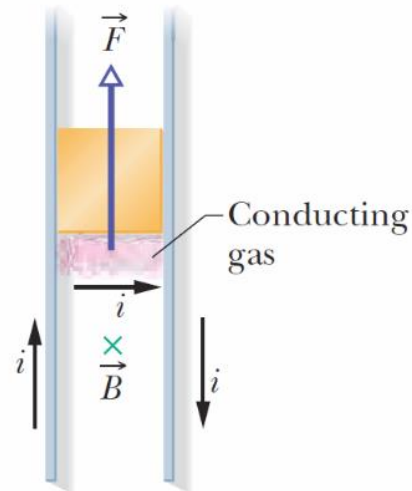
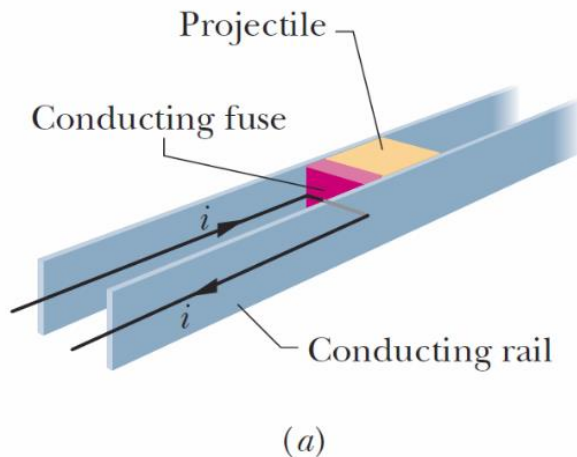
pointing towards current a .



The railgun (电磁轨道炮)

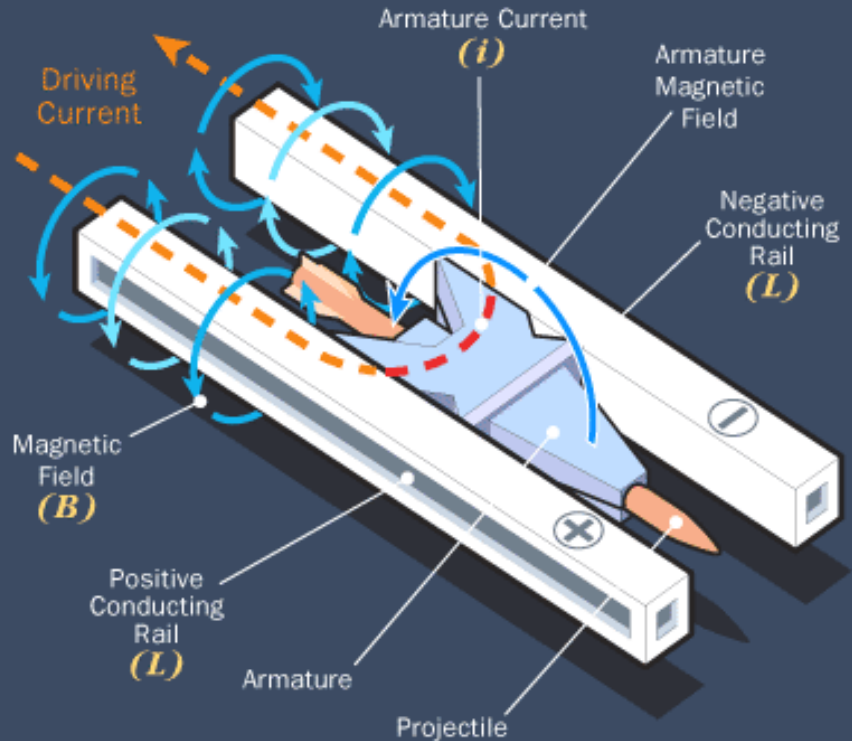
The principle of a railgun is that a current goes through the rail of width L , creating a region of conducting gas at the position shown. Then the gas pulls the bullet out of the rail.

$$F = ma = iL\langle B \rangle$$



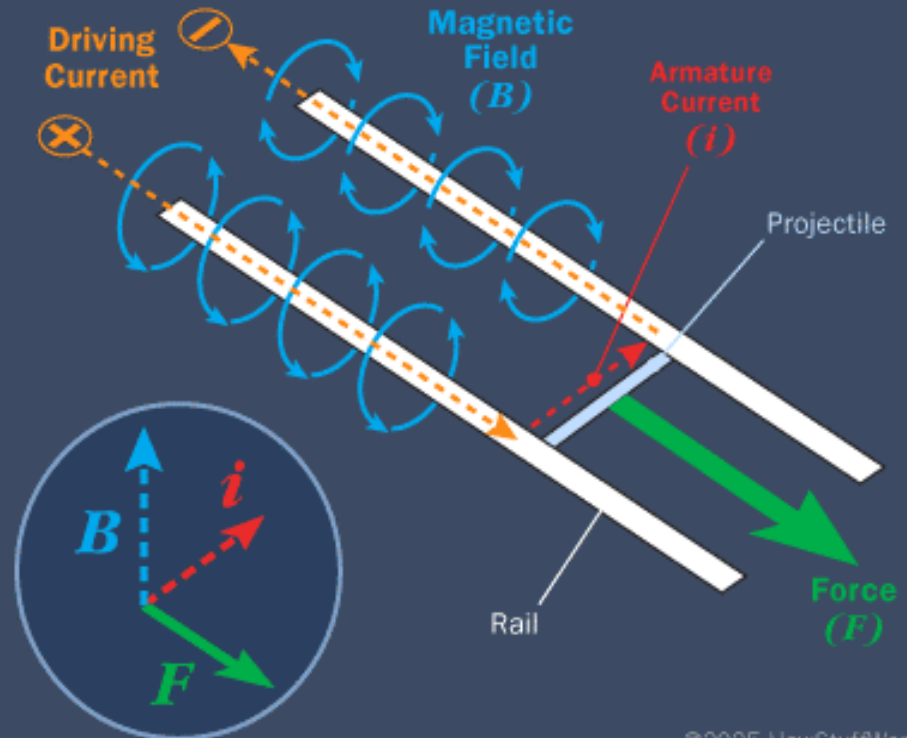
Rail gun (电磁炮)

Railguns Magnetic Field Effect



©2005 HowStuffWorks

Railguns Lorentz Force



©2005 HowStuffWorks

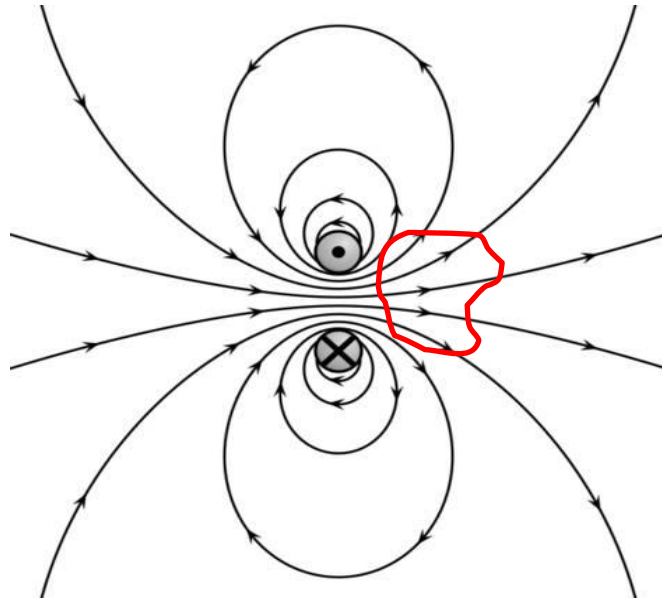
as high as 16 km /s !!!!!

Released Military Secret:

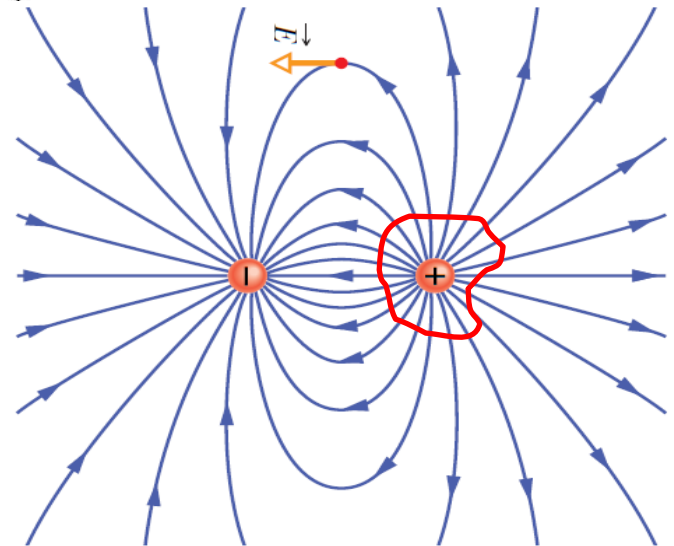
On December 10, 2010, the US Navy conducted a test of the railgun. The test set a world record by conducting a 33 MJ shot from the railgun, which was built by BAE Systems.

Gauss law for magnetism

B-field

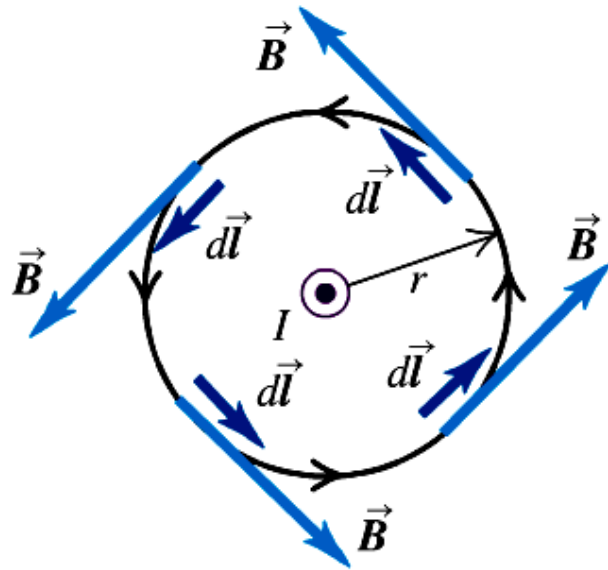


E-field



Ampere's Law

Unlike *Gauss' Law*, **Ampere's Law** is formulated not in terms of magnetic flux, but in terms of **line integral** of $\vec{B}$ around a closed path:



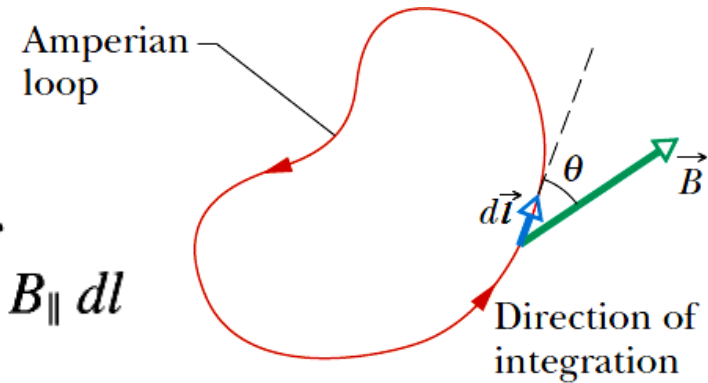
$$\oint \vec{B} \cdot d\vec{l}$$

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl$$

$$= B \oint dl$$

$$= \frac{\mu_0 I}{2\pi r} (2\pi r)$$

$$= \mu_0 I$$



A long straight wire coming out of the screen

$$B = \frac{\mu_0 I}{2\pi r}$$

Independent of the radius of the circle

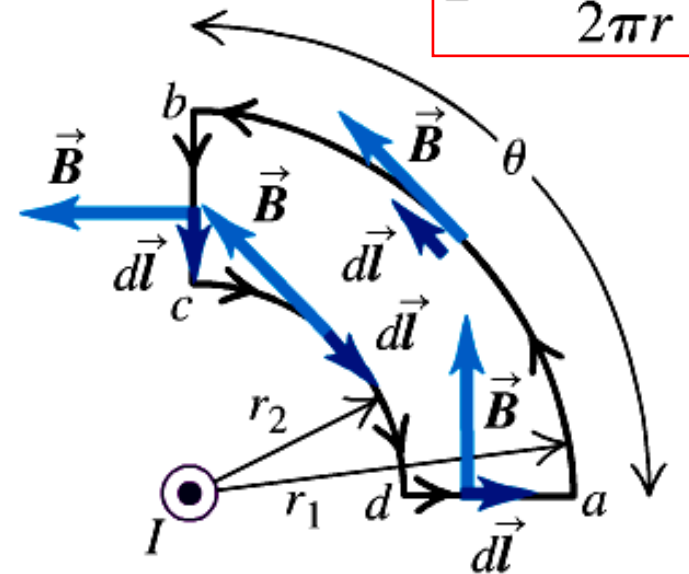
Indicating that there is a current pass through the circle

Ampere's Law

If we choose the path that **does not enclose** the conductor

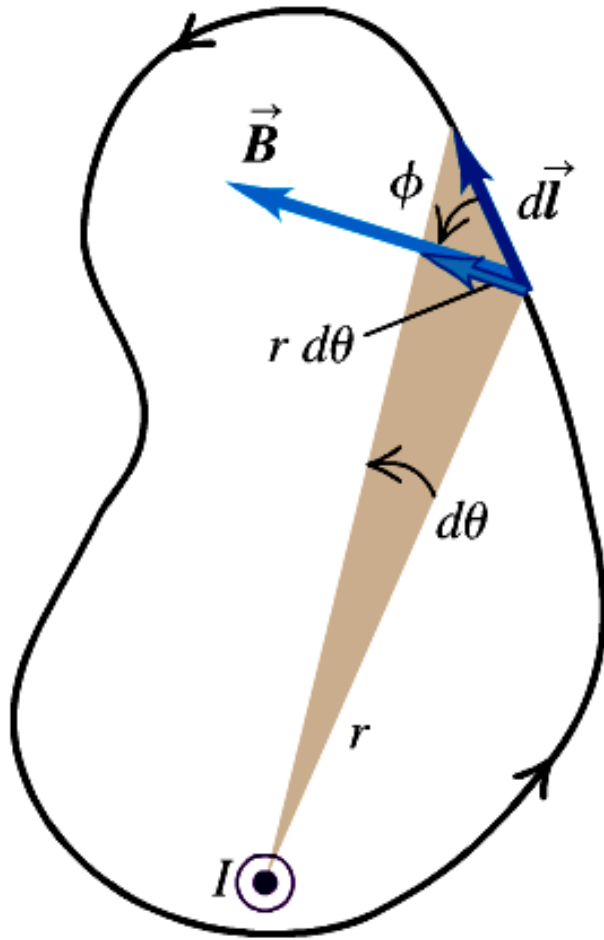
$$\begin{aligned} & \oint \vec{B} \cdot d\vec{l} \\ &= \oint B_{\parallel} dl \\ &= B_1 \int_a^b dl + (0) \int_b^c dl + (-B_2) \int_c^d dl + (0) \int_d^a dl \\ &= \frac{\mu_0 I}{2\pi r_1} (r_1 \theta) + 0 - \frac{\mu_0 I}{2\pi r_2} (r_2 \theta) + 0 = 0 \end{aligned}$$

the line integral $\oint \vec{B} \cdot d\vec{l}$ is
zero if there is no current passing through the area bounded by the path.



$$B = \frac{\mu_0 I}{2\pi r}$$

Ampere's Law – for a more general path



$$\vec{B} \cdot d\vec{l} = B dl \cos \phi$$

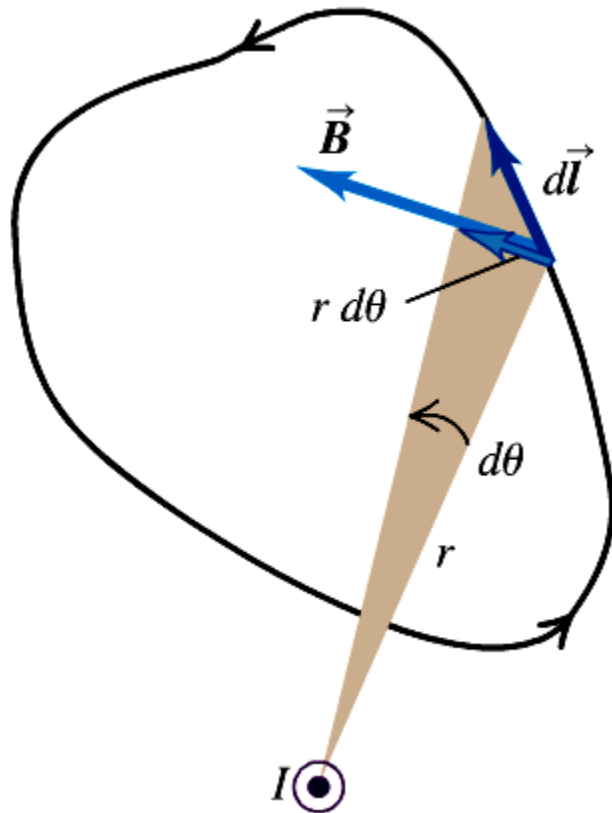
from the figure

$$dl \cos \phi = r d\theta$$

$$\begin{aligned} \therefore \oint \vec{B} \cdot d\vec{l} &= \oint \frac{\mu_0 I}{2\pi r} (r d\theta) \\ &= \frac{\mu_0 I}{2\pi} \underbrace{\oint d\theta}_{2\pi} \\ &= \mu_0 I \end{aligned}$$

Ampere's Law – for a more general path

If the path does not enclose the wire



$$\oint \vec{B} \cdot d\vec{l} = \oint \frac{\mu_0 I}{2\pi r} (r d\theta)$$

the net change in θ during the trip around the integration path is zero

$\oint d\theta$ is zero

$$\therefore \oint \vec{B} \cdot d\vec{l} = 0$$

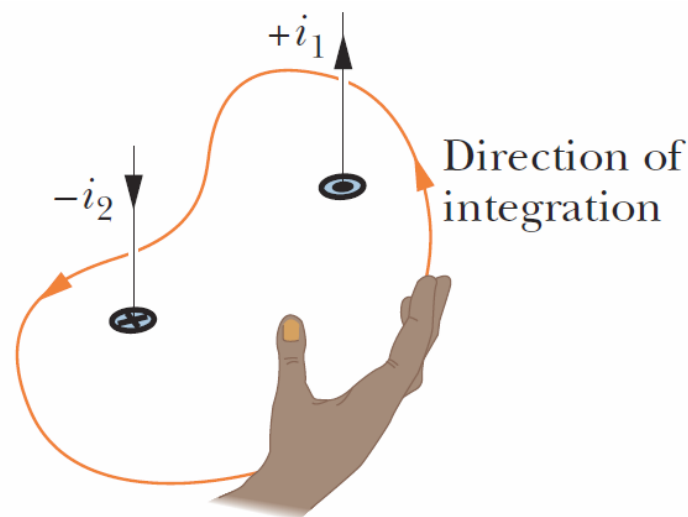
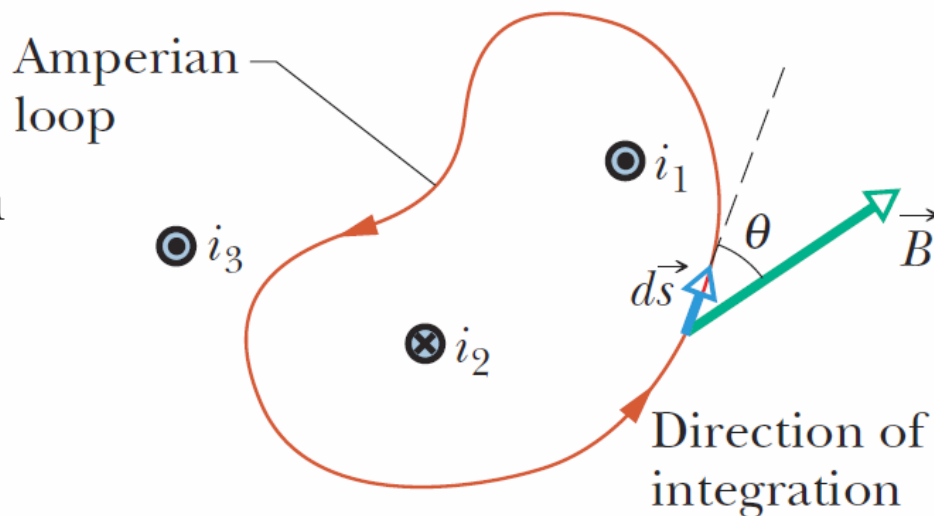
Ampere's law

Ampere's Law: the integrated magnetic-length element $\mathbf{B} \cdot d\mathbf{l}$ around a *closed* loop (Amperian loop) equals to the total current *encircled* by that loop times μ_0 .

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_{\text{enc}}.$$

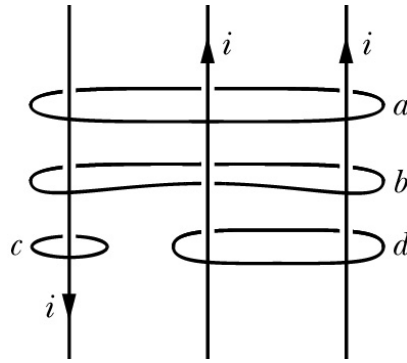
(enc – effective net current)

Signs of the encircled currents: put your right hand four fingers along the loop direction. The “plus” currents will be along the direction of your thumb.



Ampere's law

Q1:

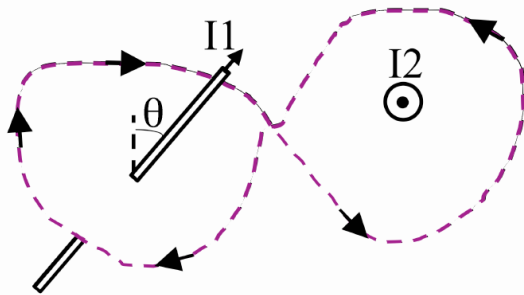


What is the value of $\oint \mathbf{B} \cdot d\mathbf{l}$ in loops a , b , c and d ?

- (a) $\mu_0 i$;
- (b) 0;
- (c) $\mu_0 i$;
- (d) $2\mu_0 i$

Q2:

What is $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{l}}$ around this purple (dashed) Amperian loop?



- A) $\mu_0 (I_2 + I_1)$
- C) $\mu_0 (I_2 + I_1 \sin\theta)$
- E) Something else!

- B) $\mu_0 (I_2 - I_1)$
- D) $\mu_0 (I_2 - I_1 \sin\theta)$



$$\mu_0 (I_1 + I_2)$$

Ampere's law in terms of current density

From the Ampere's law $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_{\text{enc}}$, remember that

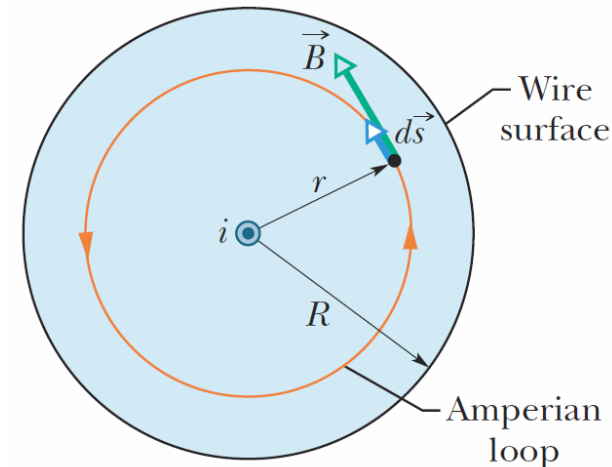
$$i_{\text{enc}} = \int_{\mathbf{A}} \mathbf{J} \cdot d\mathbf{A},$$

we get the Ampere's law in terms of the current density

$$\oint_{\partial \mathbf{A}} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_{\mathbf{A}} \mathbf{J} \cdot d\mathbf{A},$$

where $\partial \mathbf{A}$ denotes the boundary of surface $\mathbf{A}$.

This form of the Ampere's law is useful in **non-uniform current distribution** or **calculating the B -field inside a wire.**



Ampere's Law – Example 1

B-field inside a long straight wire with uniform current

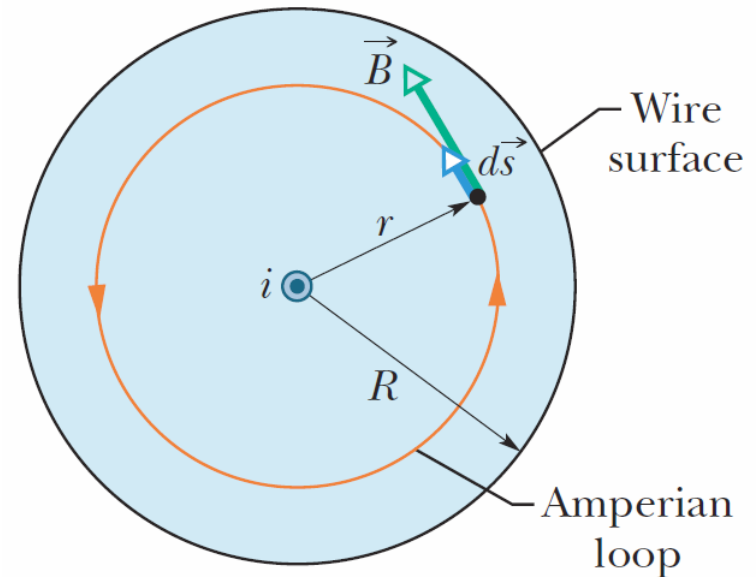
According to the figure, the current i flows out of the page. Construct a counterclockwise Amperian loop as shown.

Use Ampere's law in this loop ($\partial\mathbf{A}$):

$$\oint_{\partial\mathbf{A}} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_{\mathbf{A}} \mathbf{J} \cdot d\mathbf{A},$$

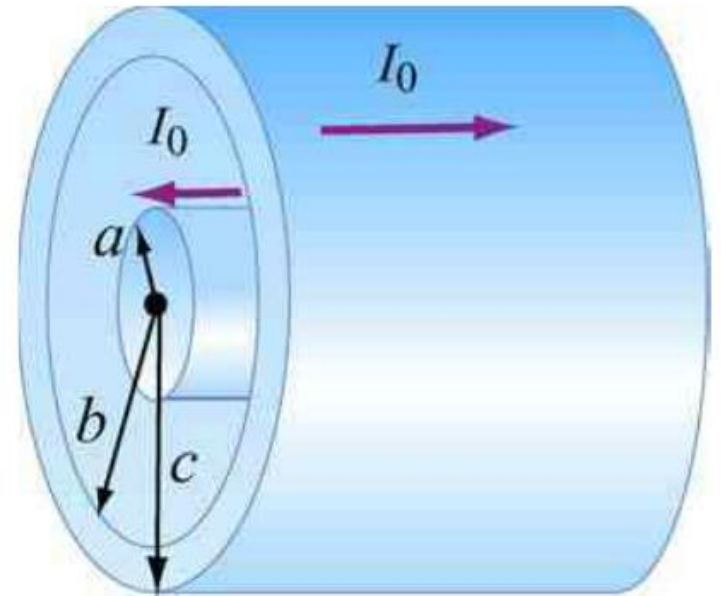
Due to symmetry, $\mathbf{B} \cdot d\mathbf{l} = Bdl$ and $\mathbf{B}$ is constant at the loop. So,

$$B \oint_{\partial\mathbf{A}} dl = \mu_0 J \int_{\mathbf{A}} dA,$$
$$B2\pi r = \mu_0 J \pi r^2, B = \frac{1}{2} \mu_0 J r.$$



Ampere's Law – Example 2

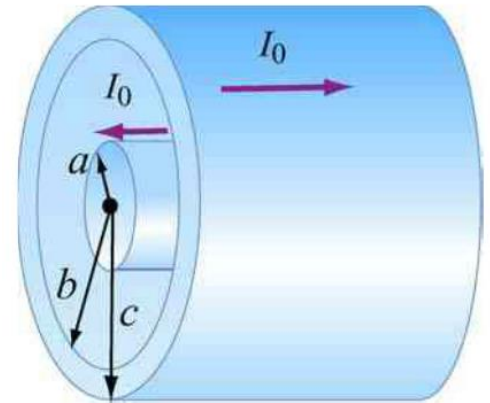
A coaxial cable consists of a solid inner conductor of radius a , surrounded by a concentric cylindrical tube of inner radius b and outer radius c . The conductors carry equal and opposite currents I_0 distributed uniformly across their cross-sections. Determine the magnitude and direction of the magnetic field at a distance r from the axis.



Ampere's Law – Example 2

There are 4 regions to be considered: $r < a$, $a < r < b$, $b < r < c$ and $r > c$. We construct a counterclockwise running, coaxial circle with radius r as the Amperian loop for all cases.

1. $r > c$. Apparently the net current inside the loop is zero. According to Ampere's law, $\mathbf{B} = 0$.
2. $b < r < c$. The net current inside the loop equals to I_0 minus a portion of the outer current,



$$I_{\text{enc}} = I_0 - \int_b^r J_{\text{outer}} dA = I_0 - \frac{I_0}{\pi(c^2 - b^2)} \pi(r^2 - b^2)$$

Ampere's Law – Example 2

According to Ampere's law,

$$\oint \mathbf{B} \cdot d\mathbf{l} = B2\pi r = \mu_0 i_{\text{enc}} = \mu_0 I_0 \left(1 - \frac{r^2 - b^2}{c^2 - b^2} \right)$$

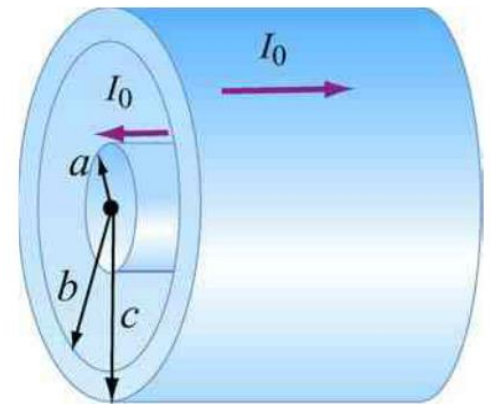
$$B = \frac{\mu_0 I_0}{2\pi r} \left(1 - \frac{r^2 - b^2}{c^2 - b^2} \right) = \frac{\mu_0 I_0}{2\pi r} \left(\frac{c^2 - r^2}{c^2 - b^2} \right).$$

The direction of $\mathbf{B}$ is counterclockwise.

3. $a < r < b$. The net current inside the loop equals to I_0 . So

$$B = \frac{\mu_0 I_0}{2\pi r},$$

pointing counterclockwise.

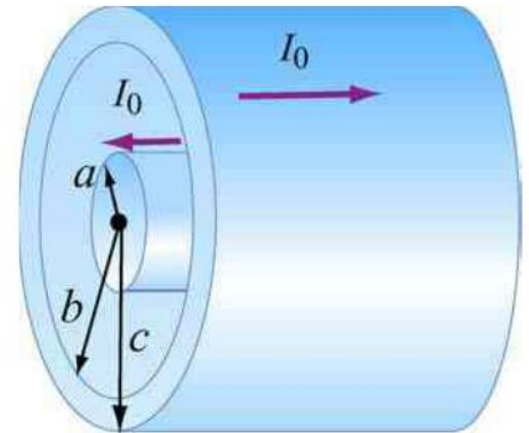
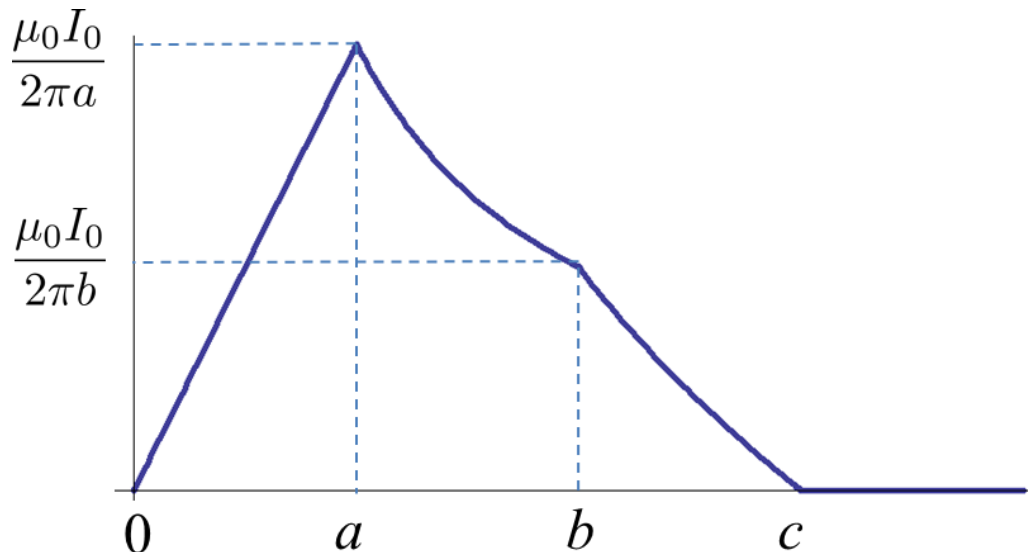


A Circulating Charged Particle

4. $r < a$. The net current inside the loop equals to

$$I_{\text{enc}} = \int_0^a J_{\text{inner}} dA = \frac{I_0}{\pi a^2} \pi r^2.$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = B 2\pi r = \mu_0 I_0 \frac{r^2}{a^2}, \quad B = \frac{\mu_0 I_0 r}{2\pi a^2}.$$



Current Sheet

An infinite flat sheet carries current out of this page.

The current is distributed uniformly along the sheet, with current per unit length given by J_S .

Find the magnetic field of this sheet.

By symmetry, $\mathbf{B}$ is $//$ to sheet & $\perp I$.

Amperian loop is a rectangle.

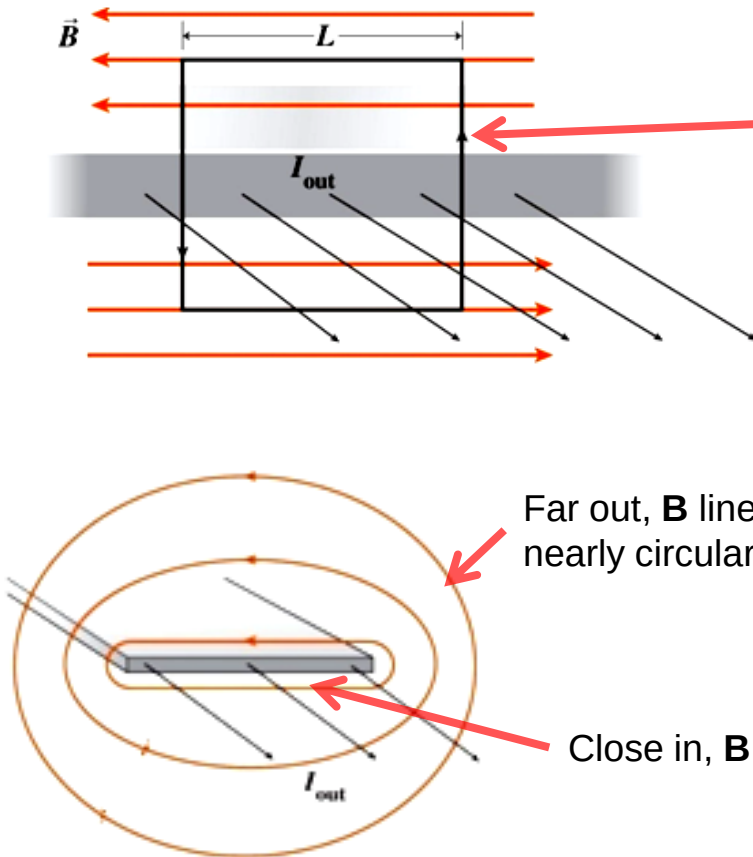
$$\oint \mathbf{B} \cdot d\mathbf{r} = 2 B L$$

$$I_{\text{enclosed}} = J_S L$$

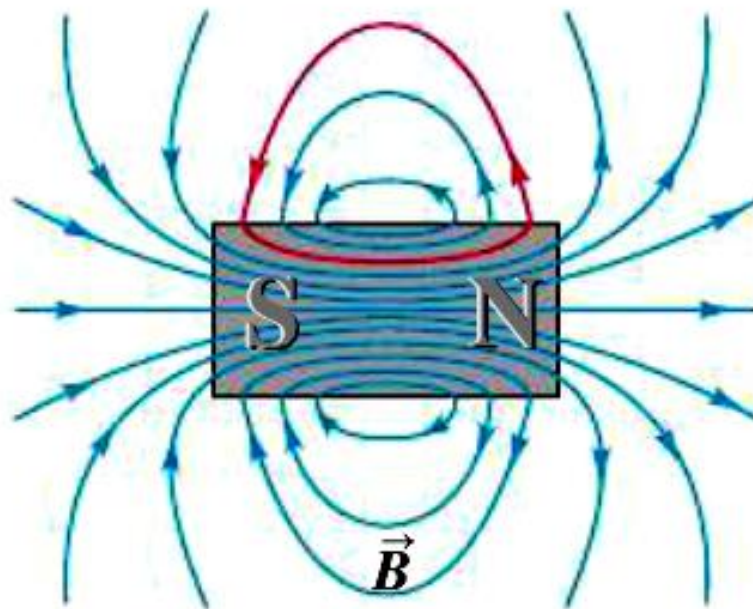
$$B = \frac{1}{2} \mu_0 J_S$$

Far out, $\mathbf{B}$ lines nearly circular

Close in, $\mathbf{B}$ lines $\sim$ infinite sheet



Open Question???



Consider the loop in red on the left passing through the permanent magnetic.

Is there a current?

Ampere's law and Gauss's Law

	Ampere's Law	Gauss's Law
Expression	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_{\text{enc}}$	$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$
Nature of integration	Loop integral	Surface integral
Infinitesimal element	$d\mathbf{l}$ (line element)	$d\mathbf{A}$ (surface element)
Coefficient	$\oint_{\partial A} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_A \mathbf{J} \cdot d\mathbf{A}$	$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{1}{\epsilon_0} \int_V \rho dV$
Density expression	μ_0	$\frac{1}{\epsilon_0}$
Differential form	$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$	$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

The Differential Form of Ampere's Law*

The integral form of the Ampere's law:
Introduce again the Nabla symbol

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_{\text{enc}}$$

$$\nabla \equiv \frac{\partial}{\partial x} \mathbf{e}_x + \frac{\partial}{\partial y} \mathbf{e}_y + \frac{\partial}{\partial z} \mathbf{e}_z$$

The mathematical curl theorem (Stokes' theorem) relates the surface integral of the curl of a vector field $\mathbf{F}$ over a surface A in Euclidean three-space to the line integral of the vector field over its boundary ∂A :

$$\int_A \nabla \times \mathbf{F} \cdot d\mathbf{A} = \oint_{\mathbf{l}=\partial A} \mathbf{F} \cdot d\mathbf{l}$$

Apply this theorem to the left of the Ampere's law, and remember that i_{enc} equals to the integral of $\mathbf{J}$ over the area of the Amperian surface, we get

$$\oint_{\mathbf{l}=\partial A} \mathbf{B} \cdot d\mathbf{l} = \int_A \nabla \times \mathbf{B} \cdot d\mathbf{A} = \mu_0 \int_A \mathbf{J} \cdot d\mathbf{A} = \mu_0 i_{\text{enc}}.$$

Getting rid of the integral yields the differential form of the Ampere's law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Magnetic Fields of Various Current Distributions

The following table lists magnetic fields caused by several current distributions. In each case the conductor is carrying current I .

Current Distribution	Point in Magnetic Field	Magnetic Field Magnitude
Long, straight conductor	Distance r from conductor	$B = \frac{\mu_0 I}{2\pi r}$
Circular loop of radius a	On axis of loop	$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$
	At center of loop	$B = \frac{\mu_0 I}{2a}$ (for N loops, multiply these expressions by N)
Long cylindrical conductor of radius R	Inside conductor, $r < R$	$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2}$
	Outside conductor, $r > R$	$B = \frac{\mu_0 I}{2\pi r}$
Long, closely wound solenoid with n turns per unit length, near its midpoint	Inside solenoid, near center	$B = \mu_0 n I$
	Outside solenoid	$B \approx 0$
Tightly wound toroidal solenoid (toroid) with N turns	Within the space enclosed by the windings, distance r from symmetry axis	$B = \frac{\mu_0 N I}{2\pi r}$
	Outside the space enclosed by the windings	$B \approx 0$

A half of the fourth Maxwell's Equation

Differential Form

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right.$$

Integral Form

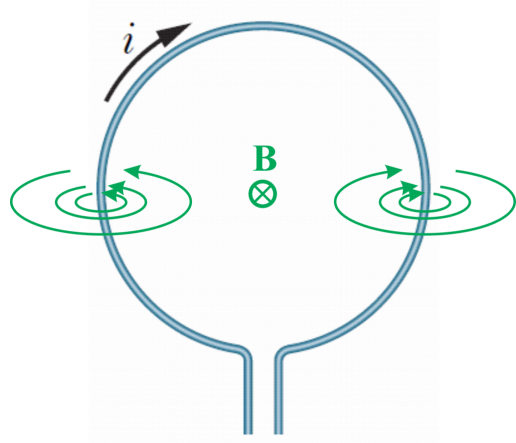
$$\left\{ \begin{array}{l} \oint_{\mathbf{A}=\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0} \\ \oint_{\mathbf{A}=\partial V} \mathbf{B} \cdot d\mathbf{A} = 0 \\ \oint_{\mathbf{l}=\partial \Sigma} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_{\Sigma} \mathbf{B} \cdot d\mathbf{\Sigma} \\ \oint_{\mathbf{l}=\partial \Sigma} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(\int_{\Sigma} \mathbf{J} \cdot d\mathbf{\Sigma} + \epsilon_0 \frac{d}{dt} \int_{\Sigma} \mathbf{E} \cdot d\mathbf{\Sigma} \right) \end{array} \right.$$

Ampere's Law

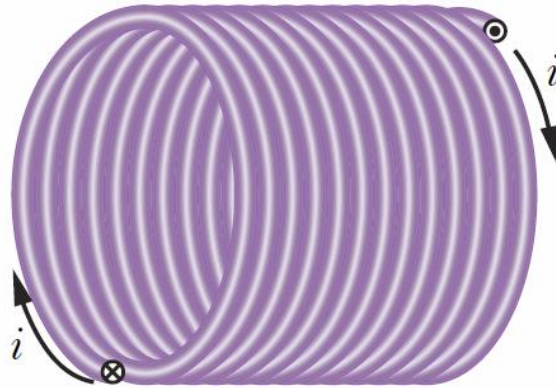


Ampere's law of the magnetic field tells us that the source for the curl of the B-field is related to the current density presented in that field. Once there's a current, there must be a vortex-like B-field around it. The direction of the vortex is the direction of the current.

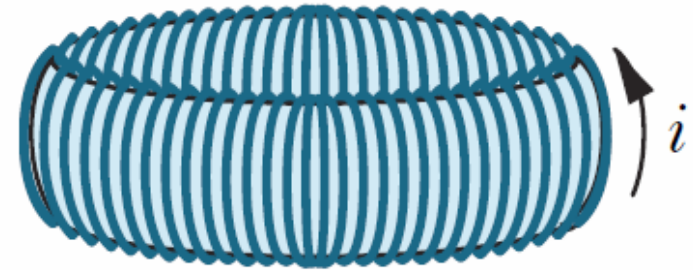
Solenoids and Toroids



A coil of wire

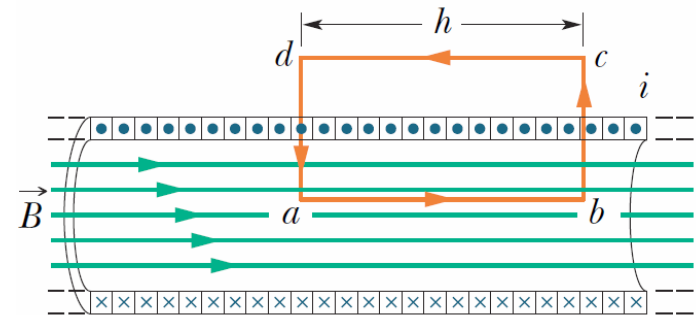
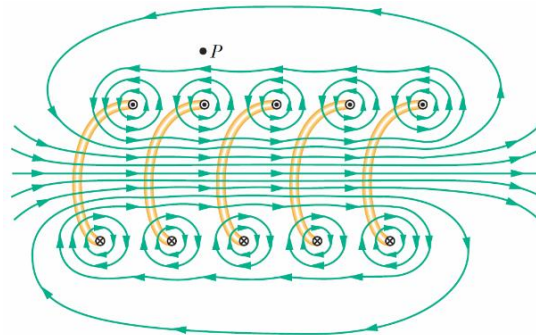
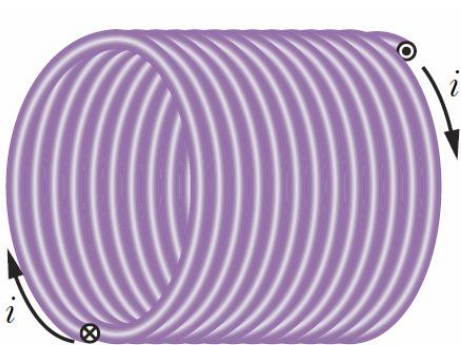


Solenoid



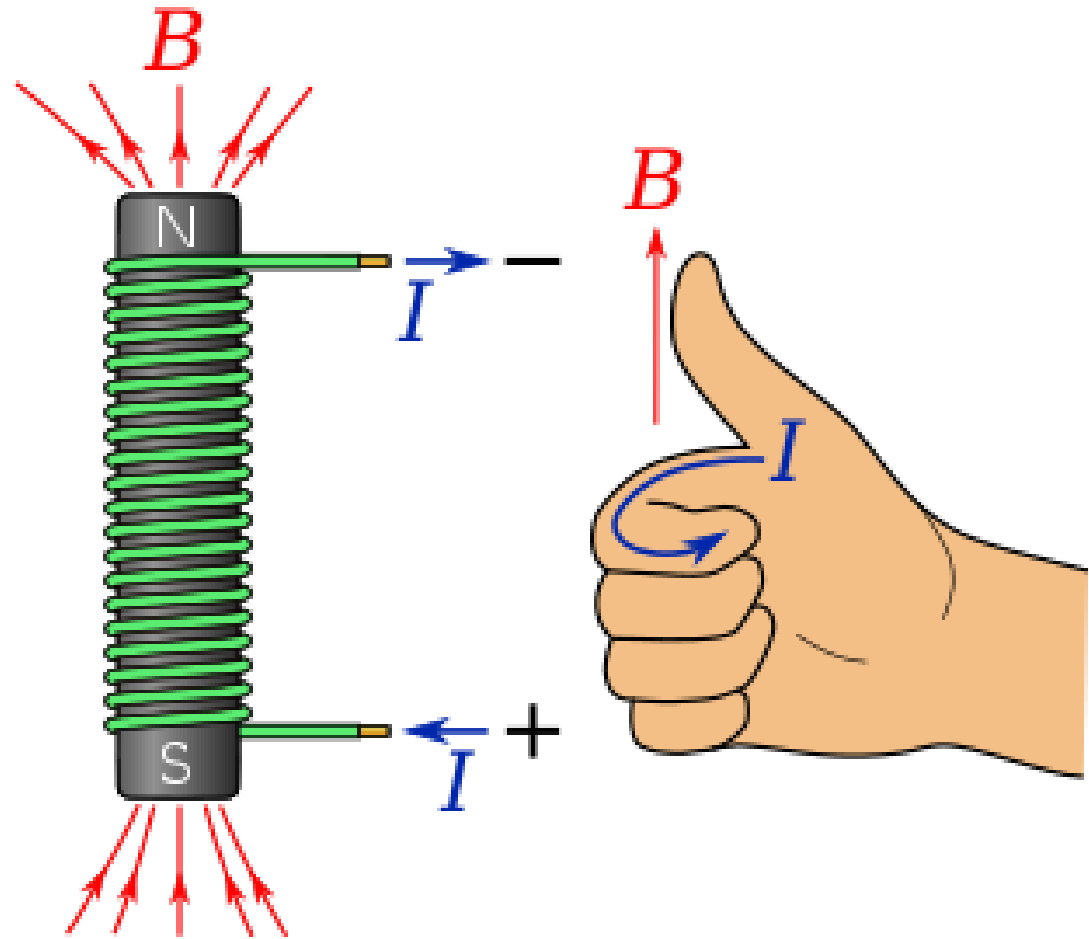
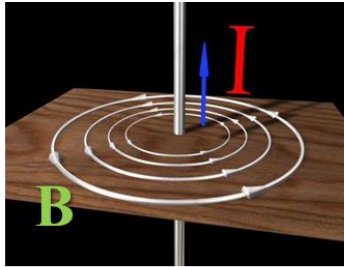
Toroid

A *solenoid* is a helical coil of wire that produces a **B**-field when current goes through. The **B**-field inside the **long** solenoid is nearly uniform.



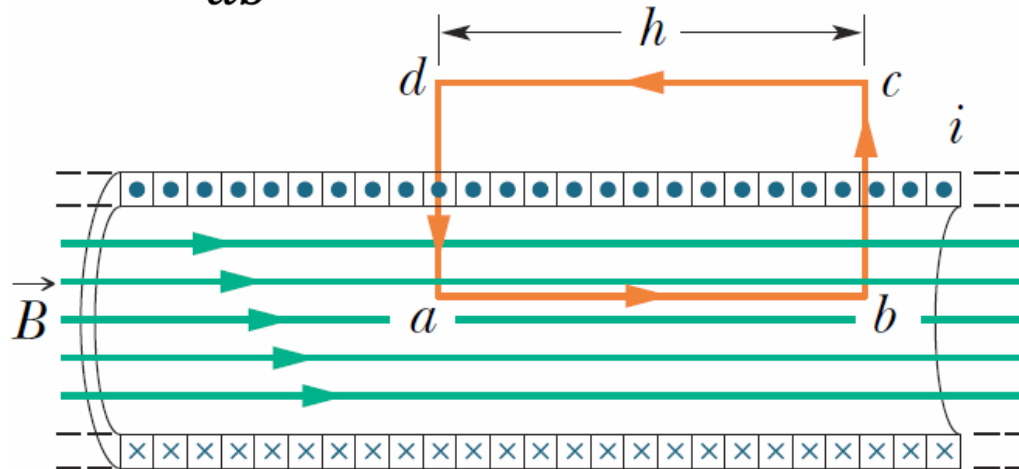
A long ideal solenoid

Solenoid- Right hand rule for direction of B



Solenoid

We apply the Ampere's law to the *ideal* solenoid on the orange Amperian loop shown. Since $\mathbf{B} = 0$ for section cd , $\mathbf{B} \cdot d\mathbf{l} = 0$ for sections bc and ad , $\oint \mathbf{B} \cdot d\mathbf{l} = Bl_{ab} = Bh$. So

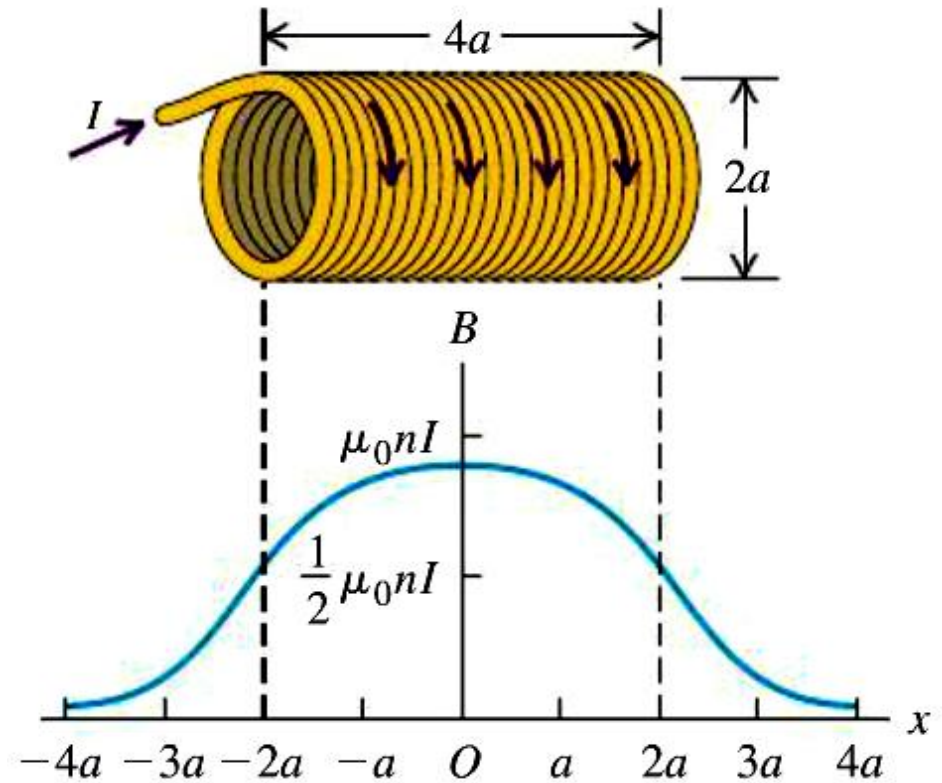
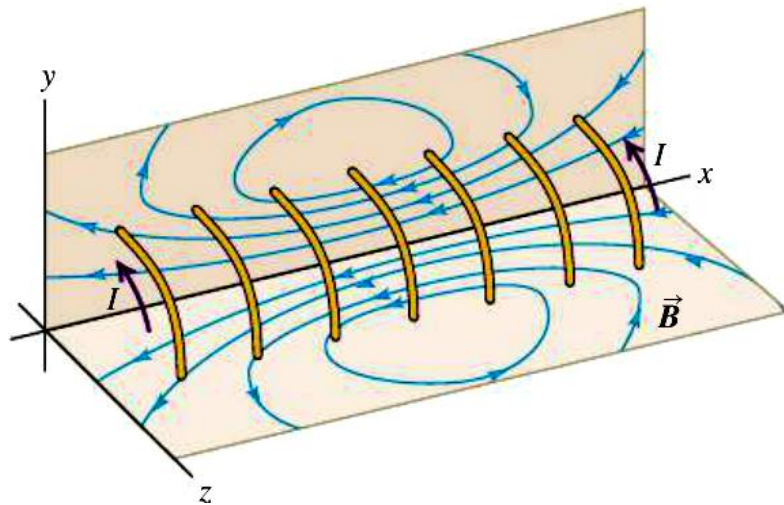
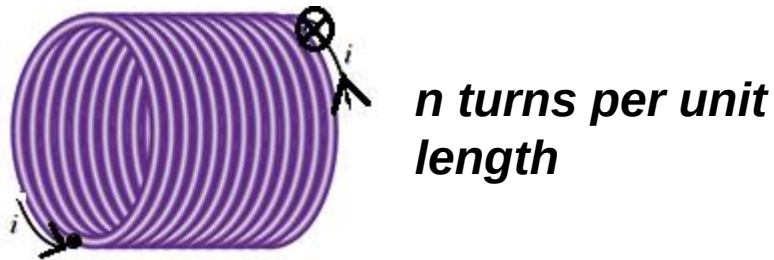


A long ideal solenoid

$$\oint \mathbf{B} \cdot d\mathbf{l} = Bh = \mu_0 i_{\text{enc}} = \mu_0 n i h, \quad B = \mu_0 n i,$$

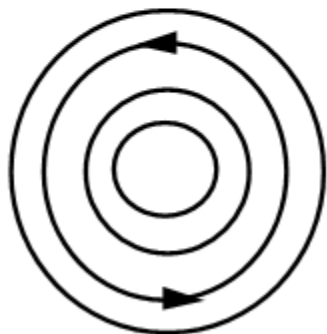
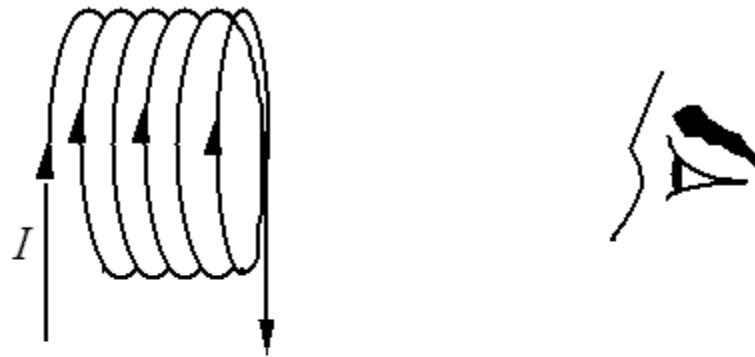
where n is the number density of the solenoid current.

Field of real solenoid

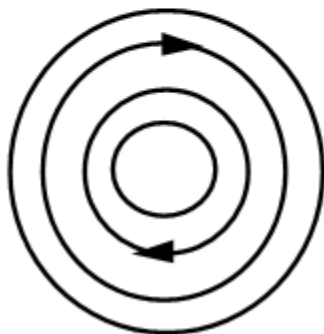


Quick Check

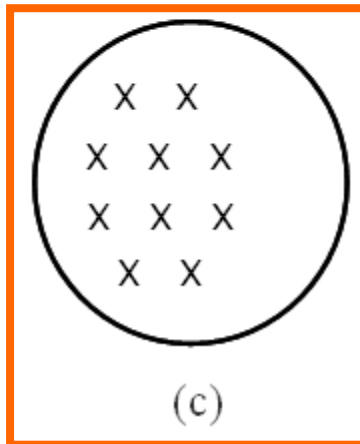
A solenoid carries current I as shown in the figure. If the observer could “see” the magnetic field inside the solenoid, how would it appear?



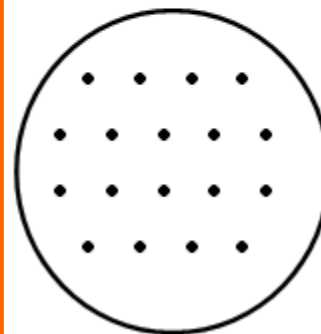
(a)



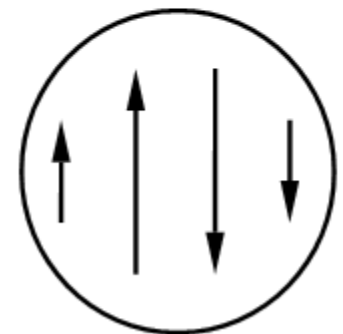
(b)



(c)



(d)



(e)

Quick Check

A wire, connected to a battery and switch, passes through the center of a long current-carrying solenoid as shown in the drawing. When the switch is closed and there is a current in the wire, what happens to the portion of the wire that runs inside of the solenoid?

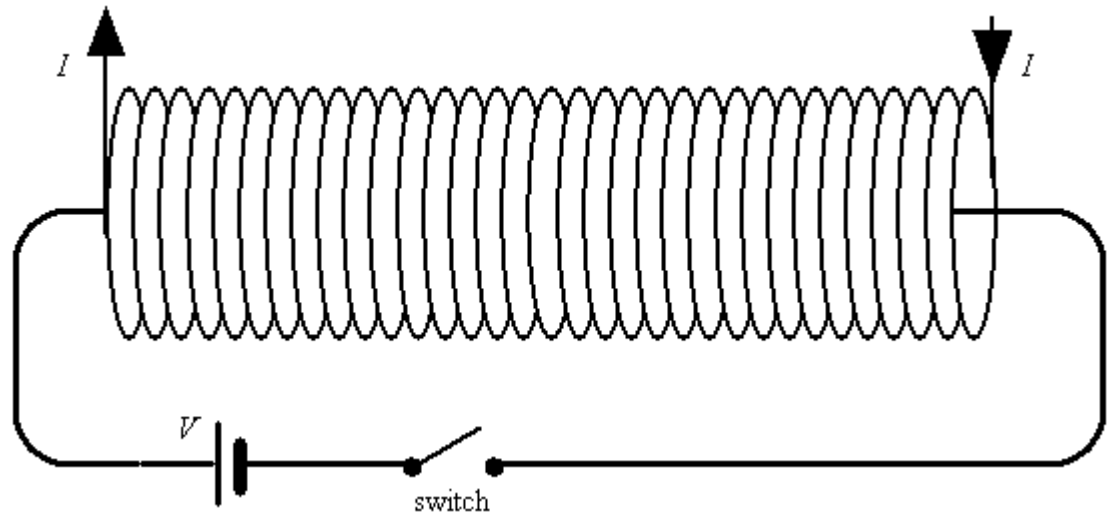
a) There is no effect on the wire.

b) The wire is pushed downward.

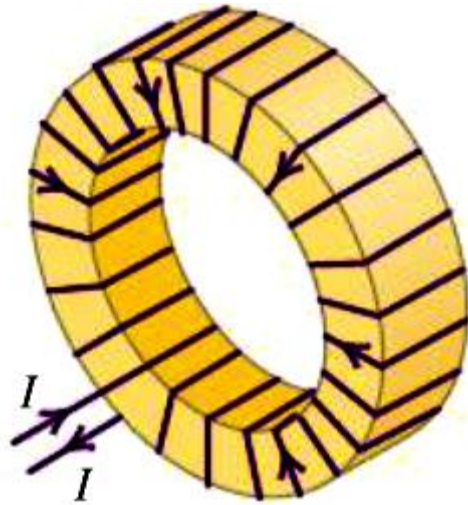
c) The wire is pushed upward.

d) The wire is pushed toward the left.

e) The wire is pushed toward the right.



Field of toroidal solenoid (toroid)



N turns

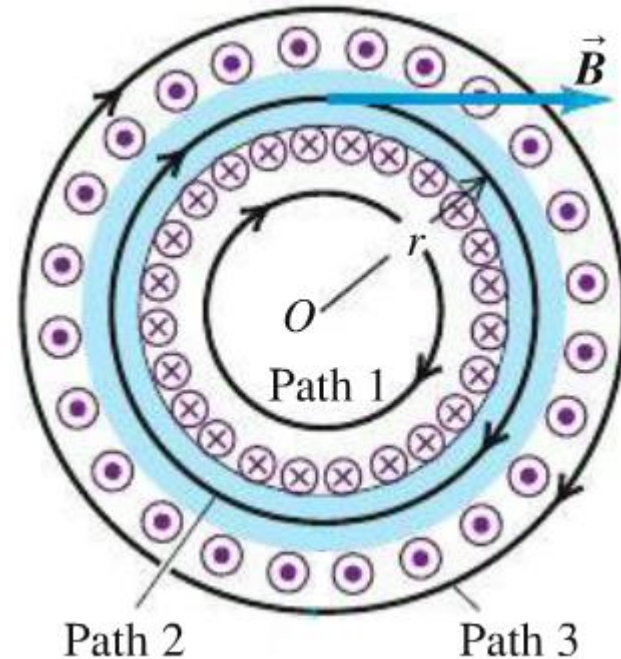
Path 1 : no current enclosed, $\mathbf{B} = \mathbf{0}$

Path 3 : net current enclosed is zero, $\mathbf{B} = \mathbf{0}$

Path 2 : $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$

$$2\pi r B = \mu_0 N I$$

$$B = \frac{\mu_0 N I}{2\pi r} \quad (\text{toroidal solenoid})$$



- $\mathbf{B}$ changes with r
- if the radial extent is small, $B = \mu_0 n I$
similar to infinitely long straight solenoid

Toroid for nuclear fusion (磁约束核聚变)

A current-carrying coil as a magnetic dipole

First we list some conclusions for this coil that we prove later.

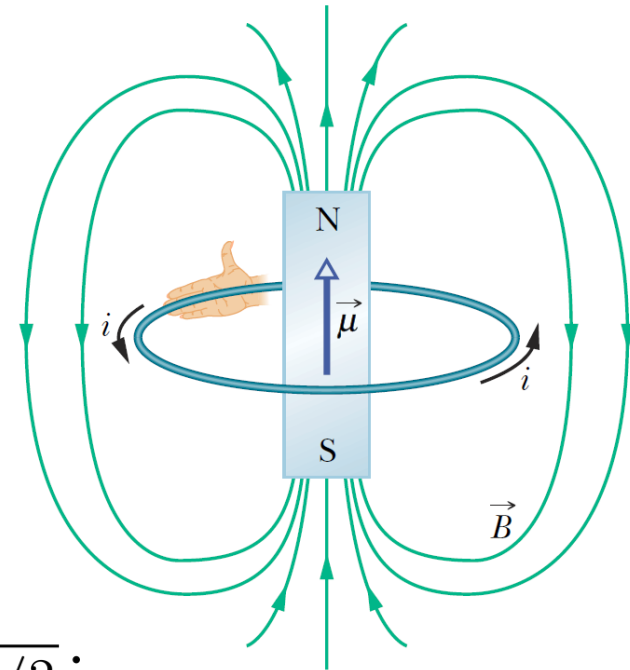
Define the magnetic dipole moment of the coil as

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B},$$
$$\boldsymbol{\mu} = Ni\mathbf{A},$$
$$U = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

where N is the number of coils, the direction of $\boldsymbol{\mu}$ and $\mathbf{A}$ (area enclosed by the coil) are determined by the right hand rule.

The $\mathbf{B}$ -field generated by this coil at point $(0, 0, z)$ is

$$\mathbf{B}(z) = \frac{\mu_0 NiR^2 \mathbf{e}_z}{2(R^2 + z^2)^{3/2}} = \frac{\mu_0}{2\pi} \frac{\boldsymbol{\mu}}{(R^2 + z^2)^{3/2}}.$$



A current-carrying coil as a magnetic dipole

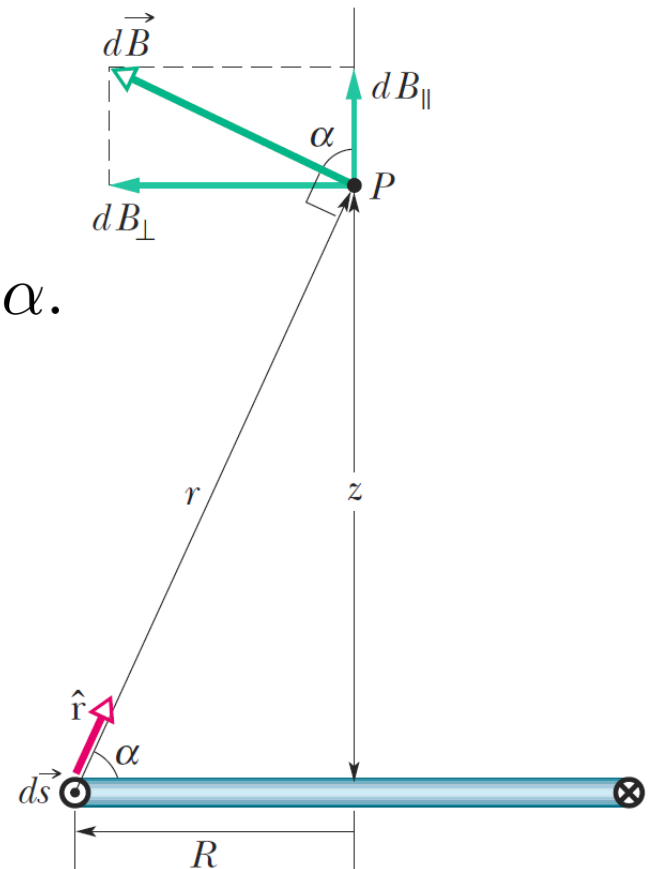
Other conclusions for a magnetic dipole still holds in this case:

Now we prove the expression of $\mathbf{B}(z)$ using
Biot-Savart Law.

$$\mathbf{B}(z) = \oint d\mathbf{B}(z) = \frac{\mu_0}{4\pi} \oint \frac{i d\mathbf{l} \times \mathbf{e}_r}{r^2} \cos \alpha.$$

In this case, $i d\mathbf{l} \perp \mathbf{e}_r$ and $r = \sqrt{R^2 + z^2}$,

$$\mathbf{B}(z) = \frac{\mu_0 i}{4\pi} \oint \frac{R d\mathbf{l} \mathbf{e}_z}{(R^2 + z^2)^{3/2}}$$



A current-carrying coil as a magnetic dipole

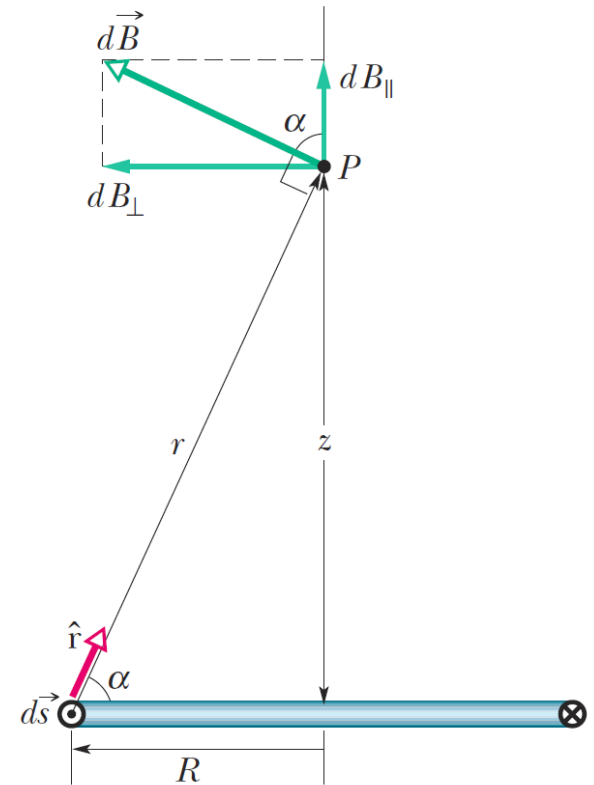
$$\mathbf{B}(z) = \frac{\mu_0 i R \mathbf{e}_z}{4\pi(R^2 + z^2)^{3/2}} \oint_l d\mathbf{l}$$
$$= \frac{\mu_0 i R^2 \mathbf{e}_z}{2(R^2 + z^2)^{3/2}}$$

If $R \ll z$,

$$\mathbf{B}(z) \approx \frac{\mu_0 i R^2 \mathbf{e}_z}{2z^3} = \frac{\mu_0 \boldsymbol{\mu}}{2\pi z^3}.$$

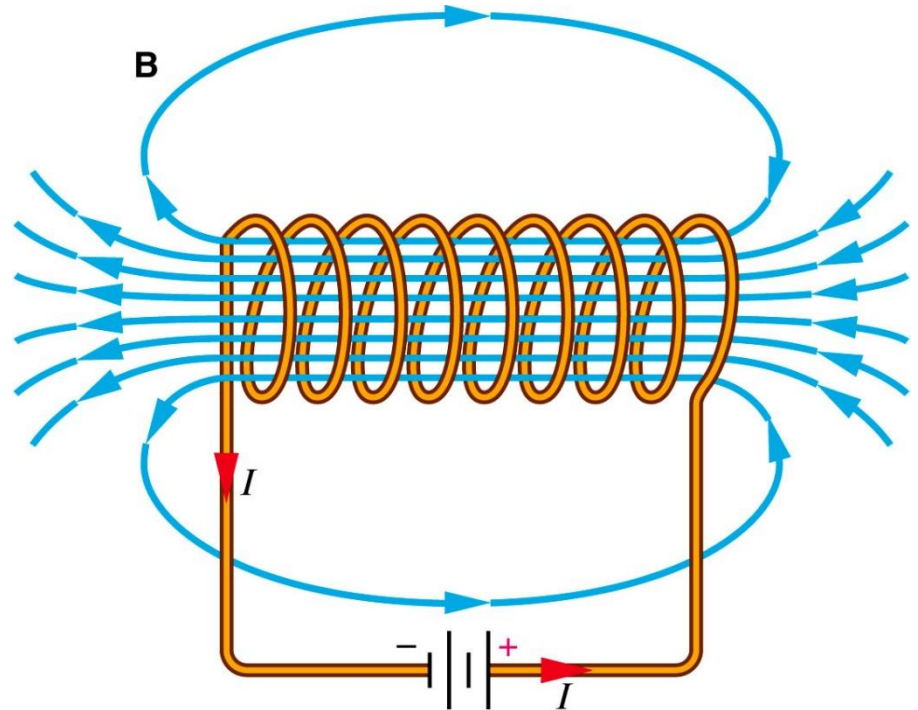
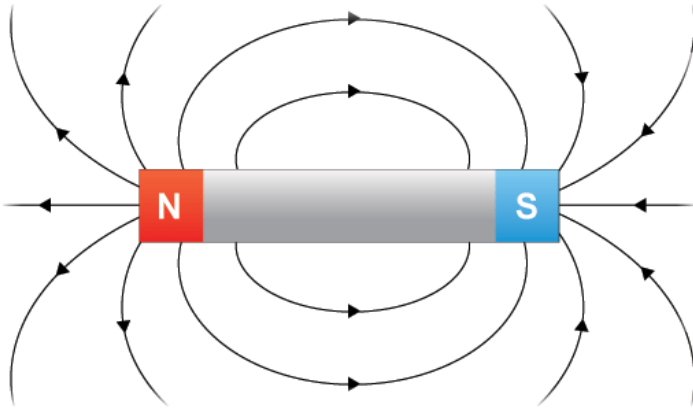
This result is analogous to the case of a far-away electric dipole,

$$\mathbf{E}(z) = \frac{\mathbf{p}}{2\pi\epsilon_0 z^3}.$$



Brainstorm

1. The magnetic field produced by a solenoid looks the same as that of a bar magnet. How to explain this?
Hint: from microscopic view of a bar magnet



Summary

Review & Summary

The Biot-Savart Law The magnetic field set up by a current-carrying conductor can be found from the *Biot-Savart law*. This law asserts that the contribution $d\vec{B}$ to the field produced by a current-length element $i d\vec{s}$ at a point P located a distance r from the current element is

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2} \quad (\text{Biot-Savart law}). \quad (29-3)$$

Here $\hat{r}$ is a unit vector that points from the element toward P . The quantity μ_0 , called the permeability constant, has the value

$$4\pi \times 10^{-7} \text{ T} \cdot \text{m/A} \approx 1.26 \times 10^{-6} \text{ T} \cdot \text{m/A}.$$

Magnetic Field of a Long Straight Wire For a long straight wire carrying a current i , the Biot-Savart law gives, for the magnitude of the magnetic field at a perpendicular distance R from the wire,

$$B = \frac{\mu_0 i}{2\pi R} \quad (\text{long straight wire}). \quad (29-4)$$

Magnetic Field of a Circular Arc The magnitude of the magnetic field at the center of a circular arc, of radius R and central angle ϕ (in radians), carrying current i , is

$$B = \frac{\mu_0 i \phi}{4\pi R} \quad (\text{at center of circular arc}). \quad (29-9)$$

Force Between Parallel Currents Parallel wires carrying currents in the same direction attract each other, whereas parallel wires carrying currents in opposite directions repel each other. The magnitude of the force on a length L of either wire is

$$F_{ba} = i_b L B_a \sin 90^\circ = \frac{\mu_0 L i_a i_b}{2\pi d}, \quad (29-13)$$

where d is the wire separation, and i_a and i_b are the currents in the wires.

Ampere's Law Ampere's law states that

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}} \quad (\text{Ampere's law}). \quad (29-14)$$

The line integral in this equation is evaluated around a closed loop called an *Amperian loop*. The current i on the right side is the *net* current encircled by the loop. For some current distributions, Eq. 29-14 is easier to use than Eq. 29-3 to calculate the magnetic field due to the currents.

Fields of a Solenoid and a Toroid Inside a *long solenoid* carrying current i , at points not near its ends, the magnitude B of the magnetic field is

$$B = \mu_0 i n \quad (\text{ideal solenoid}), \quad (29-23)$$

where n is the number of turns per unit length. Thus the internal magnetic field is uniform. Outside the solenoid, the magnetic field is approximately zero.

At a point inside a *toroid*, the magnitude B of the magnetic field is

$$B = \frac{\mu_0 i N}{2\pi} \frac{1}{r} \quad (\text{toroid}), \quad (29-24)$$

where r is the distance from the center of the toroid to the point.

Field of a Magnetic Dipole The magnetic field produced by a current-carrying coil, which is a *magnetic dipole*, at a point P located a distance z along the coil's perpendicular central axis is parallel to the axis and is given by

$$\vec{B}(z) = \frac{\mu_0}{2\pi} \frac{\vec{\mu}}{z^3}, \quad (29-27)$$

where $\vec{\mu}$ is the dipole moment of the coil. This equation applies only when z is much greater than the dimensions of the coil.